A CANONICAL MORPHISM FROM A TOTAL TENSOR-INTERNAL-HOM COMPLEX TO AN INTERNAL HOM COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Internal hom complexes and their compatibility with the monoidal structure. Remark \ref{remark-internal-hom} and Categories, Remark \ref{categories-remark-internal-hom-monoidal}, cited at proof lines 18176-18178. Produces the canonical map abstractly; the author then gives the direct construction to make the map explicit.
- Sign rules for differentials on total complexes and internal hom complexes. used at proof lines 18181-18186, 18208-18213, 18252-18257, and 18268-18275. Determine the three-term normal forms of the differentials whose signs the construction has to match.

Source reference. Original lemma 0A60: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12828–12828. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0GWQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21080–21080. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0GWJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21074–21074. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring. Given complexes $K^\bullet, L^\bullet, M^\bullet$ of R-modules there is a canonical morphism*

$$Tot(\mathrm{Hom}^\bullet(L^\bullet, M^\bullet) \otimes_R K^\bullet) \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(K^\bullet, L^\bullet), M^\bullet)$$

of complexes of R-modules functorial in all three complexes.

Proof. Via the discussion in Remark 0GWQ the existence of such a canonical map follows from Categories, Remark 0GWJ. We also give a direct construction.

Consider an element β of degree n of the right hand side. Then

$$\beta = (\beta^{p,s}) \in \prod\nolimits_{p+s=n} \mathrm{Hom}_R(\mathrm{Hom}^{-s}(K^\bullet, L^\bullet), M^p)$$

Our sign rules tell us that

$$\mathrm{d}(\beta^{p,s}) = \mathrm{d}_M \circ \beta^{p,s} - (-1)^n \beta^{p,s} \circ \mathrm{d}_{\mathrm{Hom}^\bullet(K^\bullet, L^\bullet)}$$

We can describe the last term as follows

$$(\beta^{p,s} \circ \mathrm{d}_{\mathrm{Hom}^\bullet(K^\bullet, L^\bullet)})(f) = \beta^{p,s}(\mathrm{d}_L \circ f - (-1)^{s+1} f \circ \mathrm{d}_K)$$

if $f \in \mathrm{Hom}^{-s-1}(K^\bullet, L^\bullet)$. We conclude that in some unspecified sense $\mathrm{d}(\beta^{p,s})$ is a sum of three terms with signs as follows

$$\mathrm{d}(\beta^{p,s}) = \mathrm{d}_M(\beta^{p,s}) - (-1)^n \mathrm{d}_L(\beta^{p,s}) + (-1)^{p+1} \mathrm{d}_K(\beta^{p,s}) \tag{0.2.1}$$

Next, we consider an element α of degree n of the left hand side. We can write it like so

$$\alpha = (\alpha^{t,r}) \in \bigoplus\nolimits_{t+r=n} \mathrm{Hom}^t(L^\bullet, M^\bullet) \otimes K^r$$

Each $\alpha^{t,r}$ maps to an element

$$\alpha^{t,r} \mapsto (\alpha^{p,q,r}) \in \prod\nolimits_{p+q=t} \mathrm{Hom}_R(L^{-q}, M^p) \otimes_R K^r$$

Our sign rules tell us that

$$\mathrm{d}(\alpha^{p,q,r}) = \mathrm{d}_{\mathrm{Hom}^\bullet(L^\bullet, M^\bullet)}(\alpha^{p,q,r}) + (-1)^{p+q} \mathrm{d}_K(\alpha^{p,q,r})$$

where if we further write $\alpha^{p,q,r} = \sum g_i^{p,q} \otimes k_i^r$ then we have

$$\mathrm{d}_{\mathrm{Hom}^\bullet(L^\bullet, M^\bullet)}(\alpha^{p,q,r}) = \sum (\mathrm{d}_M \circ g_i^{p,q}) \otimes k_i^r - (-1)^{p+q} \sum (g_i^{p,q} \circ \mathrm{d}_L) \otimes k_i^r$$

We conclude that in some unspecified sense $\mathrm{d}(\alpha^{p,q,r})$ is a sum of three terms with signs as follows

$$\mathrm{d}(\alpha^{p,q,r}) = \mathrm{d}_M(\alpha^{p,q,r}) - (-1)^{p+q} \mathrm{d}_L(\alpha^{p,q,r}) + (-1)^{p+q} \mathrm{d}_K(\alpha^{p,q,r}) \tag{0.2.2}$$

To define our map we will use the canonical maps

$$c_{p,q,r} : \mathrm{Hom}_R(L^{-q}, M^p) \otimes_R K^r \longrightarrow \mathrm{Hom}_R(\mathrm{Hom}_R(K^r, L^{-q}), M^p)$$

which sends $\varphi \otimes k$ to the map $\psi \mapsto \varphi(\psi(k))$. This is functorial in all three variables. With $s = q + r$ there is an inclusion

$$\mathrm{Hom}_R(\mathrm{Hom}_R(K^r, L^{-q}), M^p) \subset \mathrm{Hom}_R(\mathrm{Hom}^{-s}(K^\bullet, L^\bullet), M^p)$$

coming from the projection $\mathrm{Hom}^{-s}(K^\bullet, L^\bullet) \to \mathrm{Hom}_R(K^r, L^{-q})$. Since $\alpha^{p,q,r}$ is nonzero only for a finite number of r we see that for a given s there is only a finite number of q, r with $q + r = s$. Thus we can send α to the element β with

$$\beta^{p,s} = \sum\nolimits_{q+r=s} \epsilon_{p,q,r} c_{p,q,r}(\alpha^{p,q,r})$$

where where the sum uses the inclusions given above and where $\epsilon_{p,q,r} \in \{\pm 1\}$. Comparing signs in the equations (0.2.1) and (0.2.2) we see that

(1) $\epsilon_{p,q,r} = \epsilon_{p+1,q,r}$
(2) $-(-1)^n \epsilon_{p,q,r} = -(-1)^{p+q} \epsilon_{p,q-1,r}$ or equivalently $\epsilon_{p,q,r} = (-1)^r \epsilon_{p,q-1,r}$
(3) $(-1)^{p+1} \epsilon_{p,q,r} = (-1)^{p+q} \epsilon_{p,q,r+1}$ or equivalently $(-1)^{q+1} \epsilon_{p,q,r} = \epsilon_{p,q,r+1}$.

A good solution is to take

$$\epsilon_{p,r,s} = (-1)^{r+qr}$$

The choice of this sign is explained in the remark following the proof. □